

Computational Discrete Global Geometric Structures for Quantum Understanding

Hui Zhao

graphicsresearch@qq.com

The Lab of Computational Discrete Global Geometric Structures

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Abstract

Any particle governed by quantum mechanics, such as an electron, can be mathematically described as a type of global geometric structure on the manifold of four dimensional space-time, i.e., the whole universe. Many of the counterintuitive phenomena in quantum mechanics arise as manifestations of the global properties inherent in these geometric structures. Manipulation of physical particles to exhibit and study their global properties remains a significant challenge as it needs the whole universe as its experimental arena. However, there exist other classes of global geometric structures on two-dimensional surface, which can be computationally modeled through algorithms and visualized using computer graphics techniques. By drawing analogies between these computational global geometric structures and quantum particles, we can thus investigate and manipulate the former to gain insights into the quantum phenomena exhibited by the latter. In this paper, we visually demonstrate a specific type of global geometric structure: computational foliations, using images and videos generated by our algorithms, which are assisted by computer graphics techniques. Through this, we propose a series of analogies between the quantum mechanics of physical particles and computational foliations, including wave-particle duality, quantum self-interference, the Schrödinger equation, quantum entanglement, and quantum tunneling.

1 Introduction

The phenomena of quantum mechanics of physical particles, mathematically, can be described as some kind of global geometric structure whose arena is the whole four-dimensional space-time. Physically, we can observe these particles *locally* through all kinds of physical experiments. Apparently, it is impossible to observe them *globally* outside the whole universe.



Figure 1: Harmonic foliation on two-dimensional surface, generated by our software Geometric.

Therefore, all global properties of particles can only be indirectly reasoned about, exhibited, and derived from local phenomena. This fact of using local to understanding global leads to the anti-intuitive phenomena of quantum mechanics of physical particles.

There are many other kinds (maybe hundreds) of *global* geometric structures [Yau, 2011] in the research field of differential geometry, such as foliation (Fig. 1), differential one form (Fig. 3, holomorphic quadratic differential (Fig. 4, meromorphic quadratic differential (Fig. 2), ribbon graph (Fig. 5, spin structure, Calabi-Yau structure, and so on. Some of them could exhibit more or less similar *global* phenomena as physical particles in quantum mechanics.

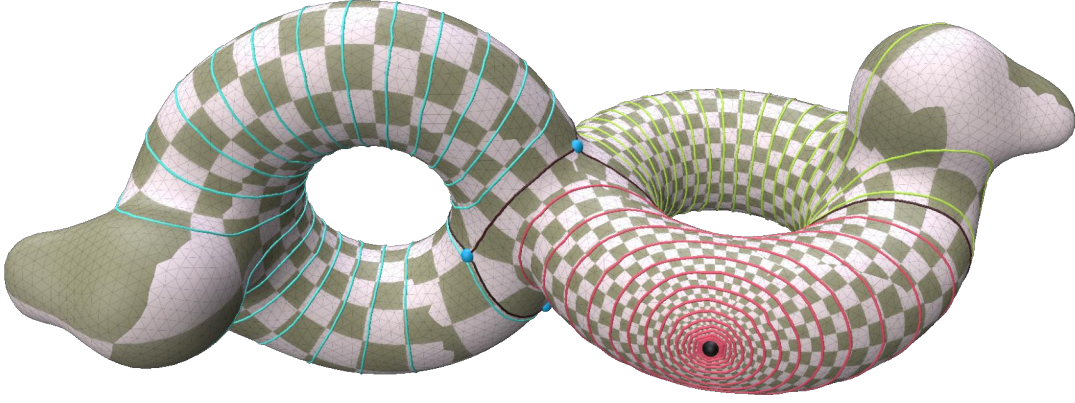


Figure 2: Meromorphic quadratic differential on a two-dimensional surface.

Every physical global geometric structure is already constructed by nature itself. On the contrary, many mathematical global geometric structures are merely proven to exist by mathematicians, with no methods for their construction. However, our understanding of quantum mechanical phenomena arises from the dynamic formation and evolutionary processes of physical particles, not merely from their existence alone.

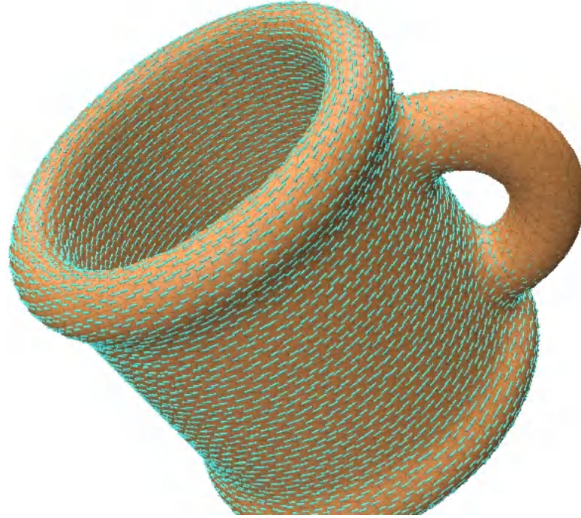


Figure 3: Differential one form on a two-dimensional surface.

A global geometric structure can not be defined by a single formula or partial differential equation (PDE) in a single coordinate, but is defined by several PDEs and their transformation relations in different coordinates. Therefore, a global geometric structure can not be constructed by numerically

calculating a single formula or solving a PDE which is expressed in a single coordinate. We need use the **algorithm** to catch their *global* property in their construction.

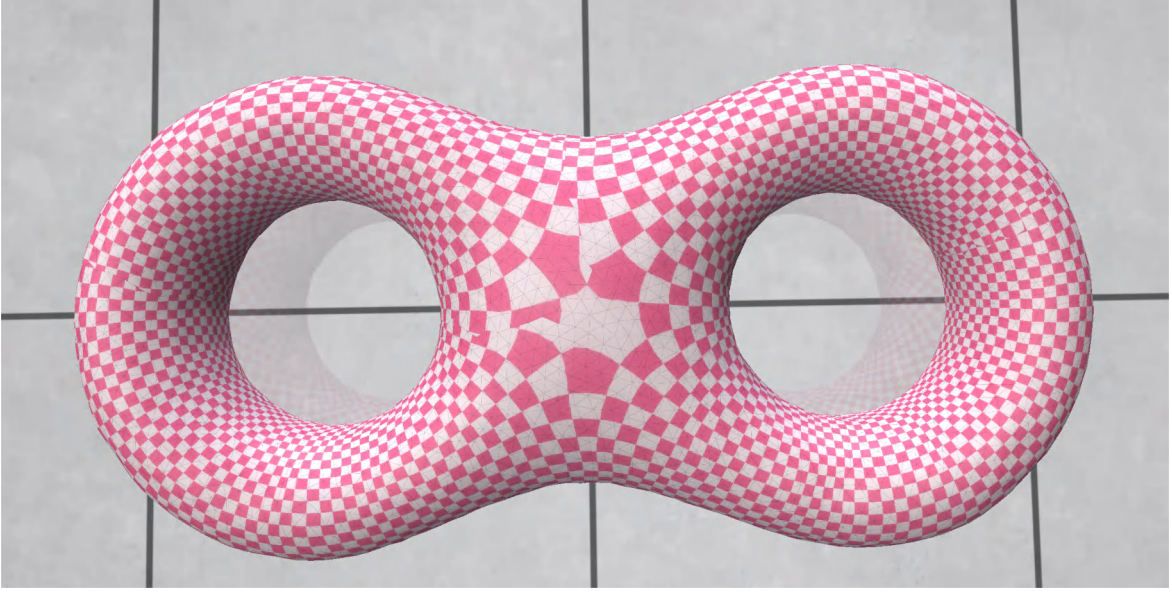


Figure 4: Quadratic differential on a two-dimensional surface.

In recent years, some mathematical global geometric structures have become constructible via novel algorithms [Zhao et al., 2022]. One such example is the harmonic foliation displayed in Fig. 1. In this paper, construction means that we can compute the numerical values of a global geometric structure by a computer and the corresponding coding.

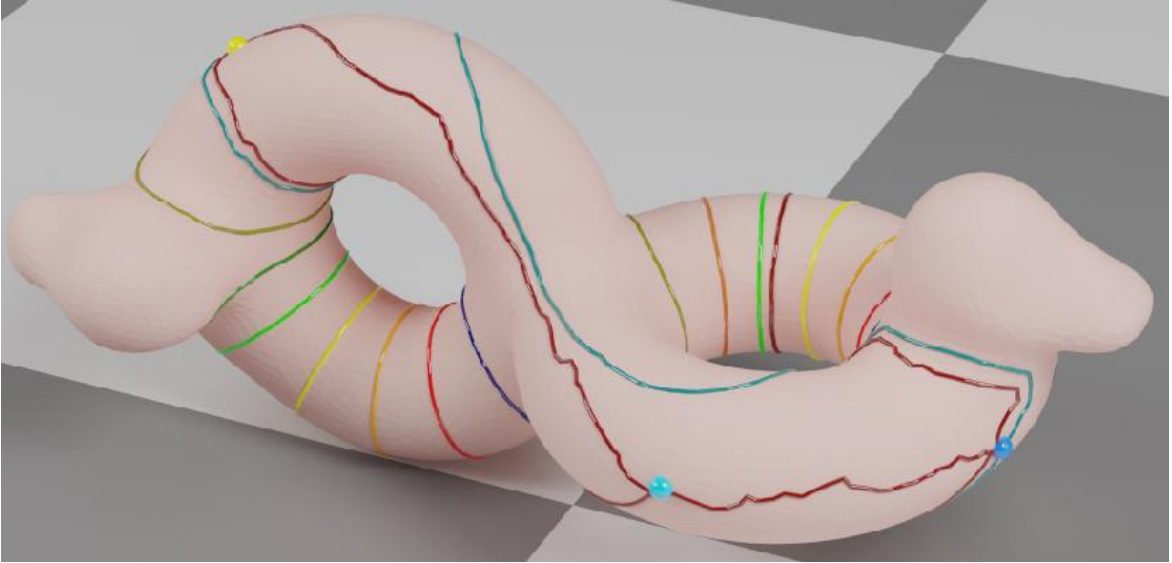


Figure 5: Ribbon graph on a two-dimensional surface.

Harmonic foliations are defined on two-dimensional surfaces. Therefore, we can observe a harmonic foliation globally from outside the surface. In contrast, physical particles living in four-dimensional space-time, there is no global visualization of them.

In this paper, our contribution lies in proposing and presenting a methodology: by drawing analogies with the global properties of other global geometric structures—specifically, harmonic foliations on surfaces—we can understand and study the global phenomena of quantum mechanics pertaining to physical particles. We demonstrate several analogies between the global features of quantum mechanics

and the global characteristics of harmonic foliations, using images and videos of harmonic foliation constructions generated by our computer algorithms on discrete meshes.

2 Computational Discrete Global Geometric Structures

2.1 Global Geometric Structures

Global geometric structures [Yau, 2011] are defined on various fiber bundles over manifolds. They are influenced by the topology of both manifolds and fiber bundles, and cannot be defined at a single point. In contrast, certain local geometric structures, such as Gaussian curvature or mean curvature, can be defined at every point.

If the topology of a manifold changes, the global geometric structure will disappear, but local geometric structure can stay unchanged. For example, the red and blue lines in Fig. 6 illustrate a global geometric structure: the holomorphic quadratic differential on a torus. If a small hole is punctured in the torus, this holomorphic quadratic differential vanishes entirely, yet the local Gaussian and mean curvatures remain unchanged in the region outside the hole.

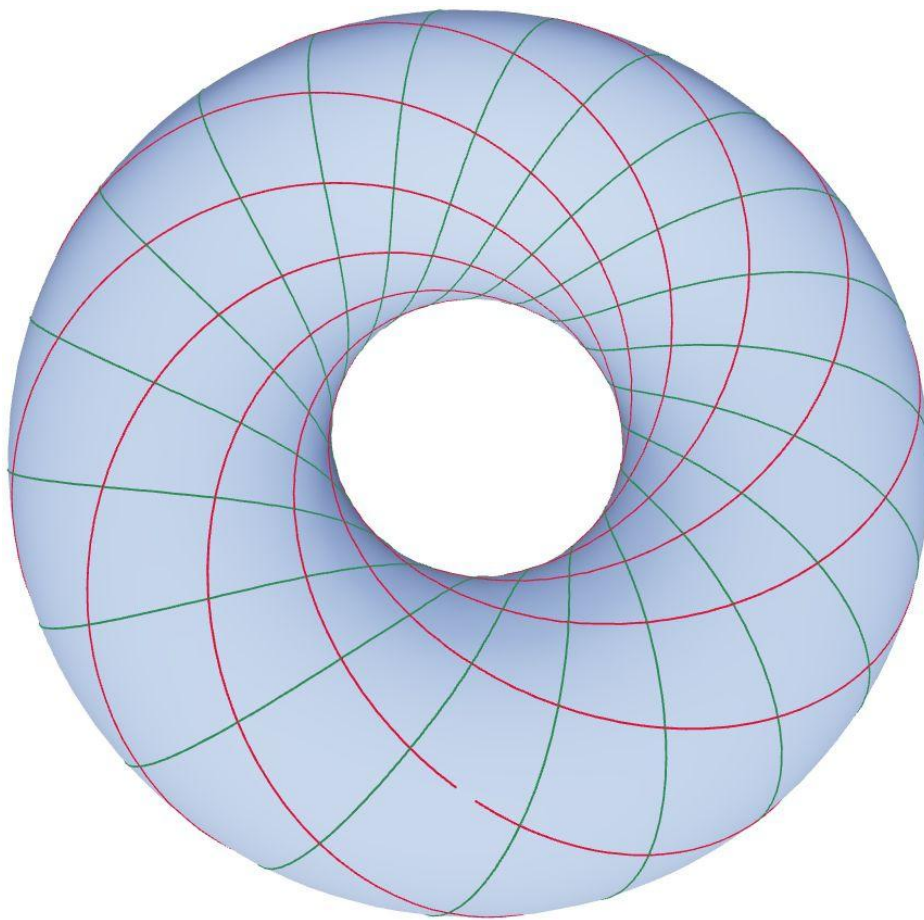


Figure 6: Holomorphic quadratic differential.

On a fiber bundle, there can be many global geometric structures. However, a global geometric structure may not exist on a given manifold or its associated fiber bundle.

Mathematically, global geometric structures still need to be defined locally via functions, tensors, or subvarieties within some local coordinate systems. And global property are encoded in their transformation rules between these local coordinate systems. For example, most geometric structures are defined by a family of special coordinate charts, where coordinate transformations preserve certain algebraic structures, such as complex structures, affine structures, or projective structures. The numerical

value of a global geometric structure at a point is different according to different local coordinates. In contrast, a local geometric structure has the same value at a point in all coordinates.

There is no single method for studying all global geometric structures; instead, researchers develop different approaches to analyze each one. Geometric analysis serves as a powerful tool for studying global geometric structures. A number of such structures are introduced and explained in [Yau, 2006] and [Yau, 2011].

2.2 Computational on Discrete Mesh

Global geometric structures are abstract objects in mathematics. In this paper, we are not studying them abstractly, but constructively and computationally. Global geometric structures can live in manifolds of any dimension. In particular, for two-dimensional surfaces, these structures can be visualized using computer graphics techniques once we successfully develop algorithms to compute them. A two-dimensional surface can be discretized into a mesh, and accordingly, smooth global geometric structures can be converted into their discrete counterparts, as shown in Fig. 7.

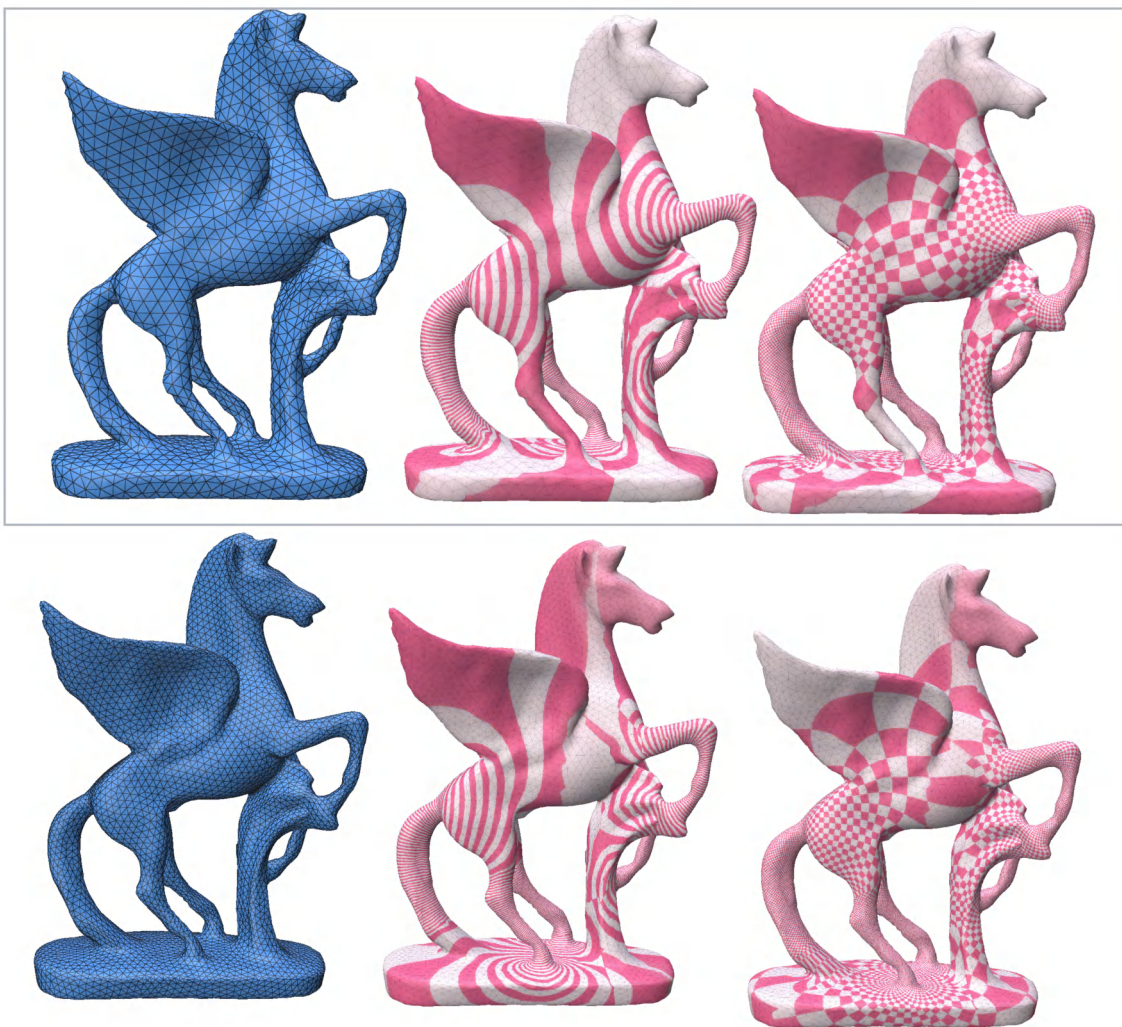


Figure 7: Global geometric structures on a two-dimensional discrete mesh.

Computing global geometric structures does not merely involve calculating a formula or solving a partial differential equation; rather, it entails designing an algorithm to capture the global properties of such structures. The reason is that although a global geometric structure can be expressed locally via a formula or PDE, it encodes global information in coordinate transformations. Consequently, calculating a local formula or solving a PDE will lose such global information. Through algorithms, we

can successfully capture their global information. We will demonstrate this using one type of global geometric structure: harmonic foliations.

Quantum particles can only be observed locally, not globally, in three-dimensional space and evolve over the time dimension. In contrast, global geometric structures such as harmonic foliations can be observed both locally and globally, as they reside on two-dimensional surfaces—spaces that can be fully visualized within three-dimensional space.

3 Harmonic Foliations

In this section, we provide a brief overview of the mathematical background pertaining to harmonic foliations and other related mathematical definitions. For further details, refer to [Strebel, 1984; Hubbard and Masur, 1979; Thurston and Levy, 1997; Lei et al., 2017a,b; Crane et al., 2013; Fortier Bourque, 2018; Jost, 2006; Palmer, 2015; Gortler and Thurston, 2015].

3.1 Complex Structure

A 2-dimensional manifold is locally expressed as $\mathbb{R}^2$. It is represented by a collection of charts $\{(U_i, \phi_i)\}_{i=1}^2$ with some overlapping regions. $\phi_i : U_i \rightarrow \mathbb{R}^2$ denotes a bijective map from U_i to an open set $\phi_i(U_i)$. On the overlap of two charts (U_i, ϕ_i) and (U_j, ϕ_j) , the *transition function* $\phi_{ij} = \phi_j \circ \phi_i^{-1} : \phi_i(U_i \cap U_j) \rightarrow \phi_j(U_j \cap U_j)$ describes a bijective transformation from $\phi_i(U_i)$ to $\phi_j(U_j)$. The set of all charts $\{(U_i, \phi_i)\}_{i=1}^2$ forms an atlas.

The charts and their transition functions determine a global geometric structures. Different kinds of transition functions will produce different global geometric structures.

A complex function $f : \mathbb{C} \rightarrow \mathbb{C} : x + iy \mapsto u(x, y) + iv(x, y)$ is said to be *holomorphic*, if it satisfies the *Cauchy-Riemann* equation $u_x = v_y, u_y = -v_x$. If f is invertible and f^{-1} is also holomorphic, then f is called a *biholomorphic function*. If all the transition functions of an atlas $\{(U_i, \phi_i)\}_{i=1}^2$ are biholomorphic, then the atlas is called a *complex atlas*. The maximal complex atlas is termed a *complex structure*. A surface equipped with a complex structure is known as a *Riemann surface* [Jost, 2006]. In the context of Riemann surfaces, a complex structure also constitutes a conformal structure.

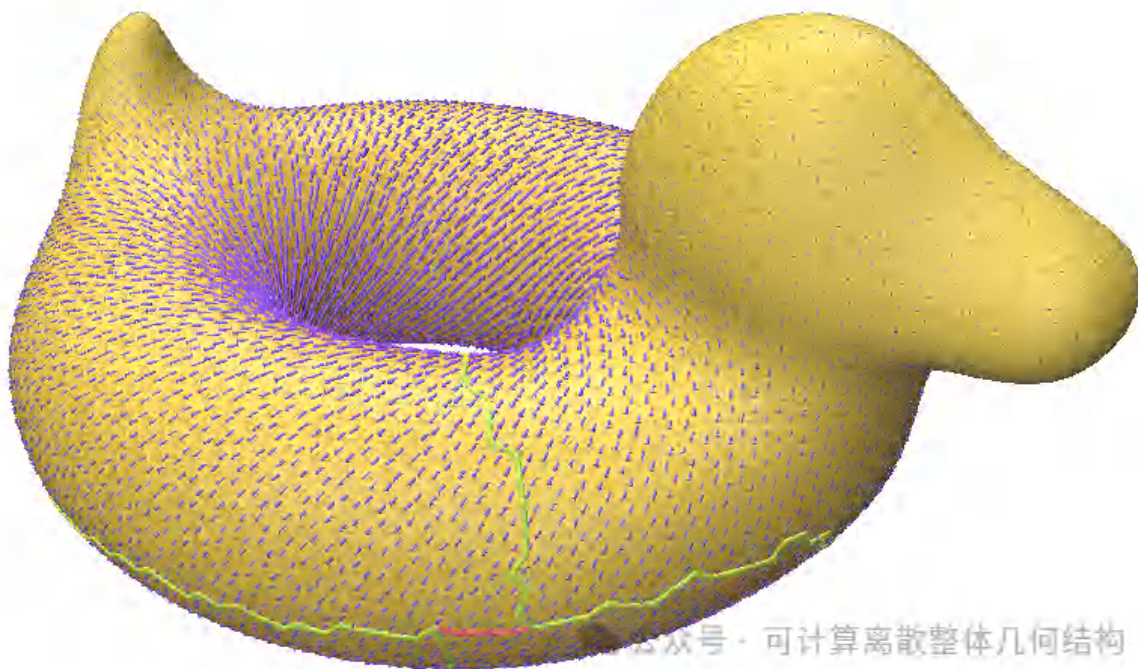


Figure 8: A differential on a two-dimensional 1-form, generated by software Geometric.

3.2 Differential 1-Form

On a Riemann surface R equipped with a complex atlas $\{(U_\alpha, \phi_\alpha)\}$, a complex-valued differential one-form $\omega : TR \rightarrow \mathbb{C}$ maps any tangent vector $v \in TR$ and scalar $\lambda \in \mathbb{C}$ to a complex number in $\mathbb{C}$ [Palmer, 2015], satisfying the following properties:

$$\omega(\lambda v) = \lambda \omega(v).$$

Locally, ω is defined by a family $\{(U_\alpha, \phi_\alpha, \omega_\alpha)\}$, such that on each local chart with the local complex parameter z_α , we have $\omega_\alpha = \omega_\alpha(z_\alpha)dz_\alpha$. On the overlap $U_\alpha \cap U_\beta$ of two charts, where $z_\alpha = \phi_{\alpha\beta}(z_\beta)$ denotes the coordinate transformation, then coefficient function ω_α transforms according to the rule $\omega_\alpha(z_\alpha) \frac{dz_\alpha}{dz_\beta} = \omega_\beta(z_\beta)$. If $\omega_\alpha(z_\alpha)$ is a holomorphic function on every chart, ω is referred to as a *holomorphic differential 1-form*. A point $p \in R$ where $\omega_\alpha(p) = 0$ is called a *zero* of ω . this property is chart-independent. A illustration of differential 1-form is in Fig. 8.

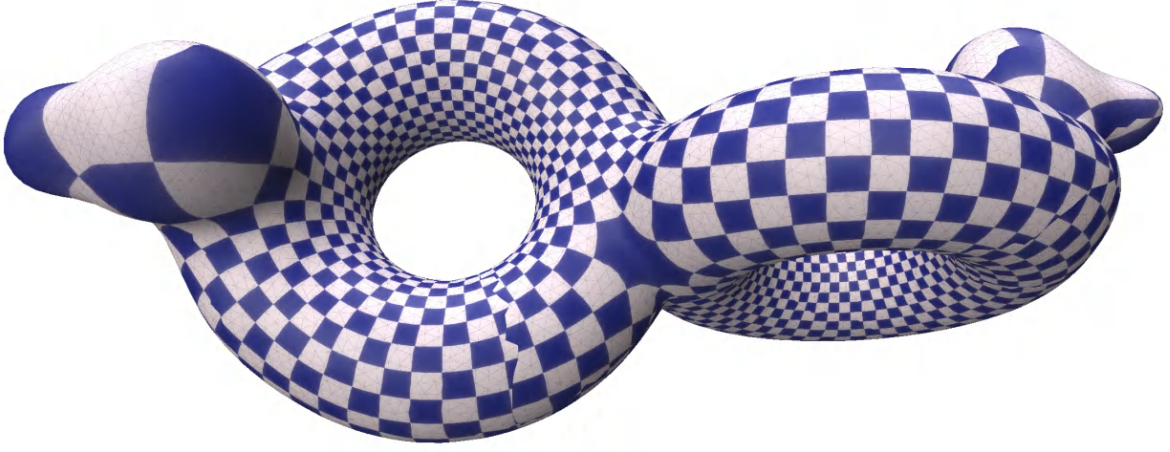


Figure 9: A holomorphic quadratic differential on a two-dimensional discrete mesh, generated by software Geometric.

3.3 Holomorphic Quadratic Differential

Differential 1-form ω maps any tangent vector to a value *linearly*, whereas a quadratic differential $\Omega : TR \rightarrow \mathbb{C}$ *quadratically* maps any tangent vector λv quadratically [Palmer, 2015], specifically satisfying:

$$\Omega(\lambda v) = \lambda^2 \Omega(v).$$

Quadratic differentials can be regarded as “*locally* the square of differential 1-form”. On a Riemann surface R , given a coordinate chart $z_\alpha : U_\alpha \subset R \rightarrow \mathbb{C}$, a quadratic differential Ω defined on the entire Riemann surface R can be expressed locally on U_α as:

$$\Omega_\alpha = \Omega_\alpha(z_\alpha)dz_\alpha^2,$$

Here, $\Omega_\alpha(z_\alpha)$ is a holomorphic function on each chart, and the map Ω is then referred to as a *holomorphic quadratic differential*. Between different coordinate charts, the coefficient function $\Omega_\alpha(z_\alpha)$ transforms according to the following rule:

$$\Omega_\alpha(z_\alpha) \frac{dz_\alpha^2}{dz_\beta^2} = \Omega_\beta(z_\beta).$$

If a holomorphic quadratic differential has poles, it is called as meromorphic quadratic differential. A visual exhibition of holomorphic quadratic differential is shown in Fig. 9 and a meromorphic quadratic differential is shown in Fig. 10 and Fig. 11.

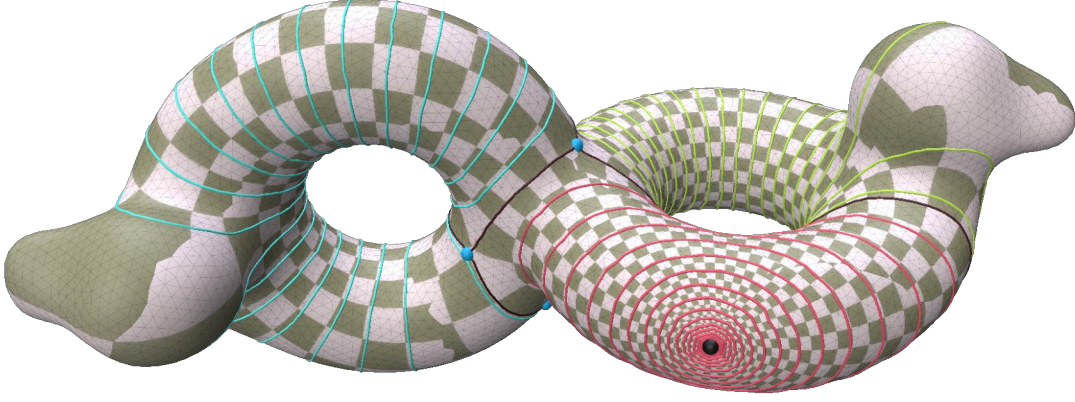


Figure 10: A foliation with pole and its corresponding meromorphic quadratic differential on a two-dimensional discrete mesh, generated by software Geometric.

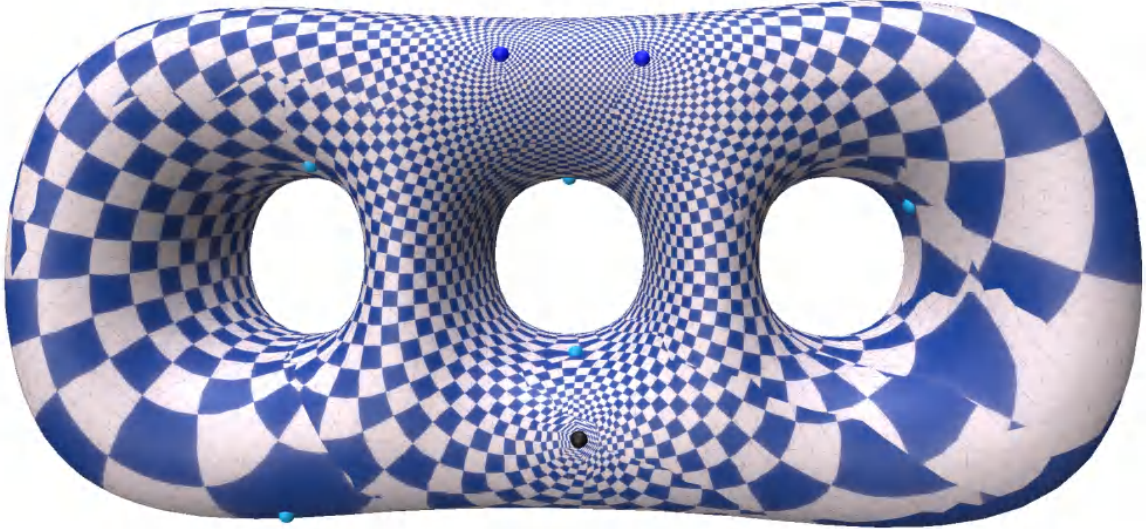


Figure 11: A meromorphic quadratic differential on a two-dimensional discrete mesh, generated by software Geometric.

Definition 3.1 (Zero). A point p is called a *zero* of a holomorphic quadratic differential if $\Omega_\alpha(p) = 0$. Such zeros are chart-independent, as shown by green points in Fig.10.

Definition 3.2 (Natural Coordinate). On a simply-connected patch U_i containing no *zeros*, the holomorphic square root of Ω is well-defined, equivalently, it takes the form of a holomorphic differential 1-form: $\sqrt{\Omega(p)}dz$. A holomorphic function $\zeta : U_i \rightarrow \mathbb{C}$ can be defined via the path-independent integral $\zeta(p) = \int_{p_0}^p \sqrt{\Omega(z)}dz$ where the integral is taken from any point p to a fixed point p_0 . This function ζ is referred to as a *natural coordinate* on U_i .

The natural coordinate induced by a quadratic differential gives rise to a flat metric with a conical angle deficit of $-\pi$ at *zeros*. For a quadratic differential, a vector $v \in T_p R$ at any point p on the Riemann surface R , is termed *horizontal* if $\Omega(v)$ is real and positive. Away from *zeros*, the pullback of the positive real axis under Ω forms a line (rather than a vector). Curves whose tangent vectors are horizontal are called the *horizontal trajectories* of Ω . Vertical (resp. horizontal) trajectories are curves along which the natural coordinate remains constant in its real (resp. imaginary) part. Trajectories passing through zeros are referred to as *critical trajectories*, as show by black lines in Fig. 13.

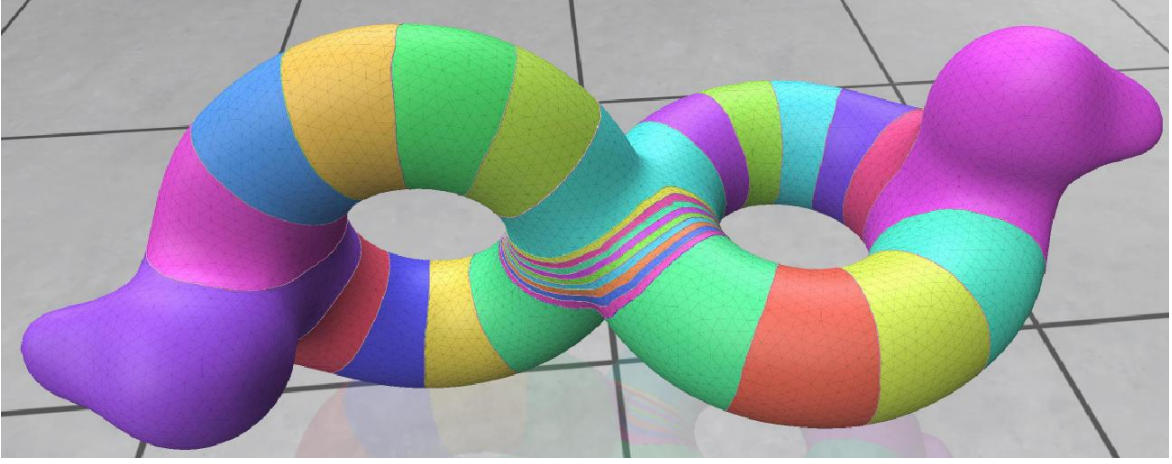


Figure 12: A non-harmonic foliation on a two-dimensional discrete mesh, generated by software Geometric.

3.4 Measured Foliation

A foliation [Fathi, 2012] resembles stripe patterns on a surface (see Fig. 12 and Fig. 13). A foliation could be harmonic or non-harmonic. A *leaf* is simply a smooth curve (closed or not) on the surface. A measure μ can be associated with a foliation $\mathcal{F}$, representing the number of leaves crossed by a transverse arc γ . A measured foliation $|dv|$ or $((\mathcal{F}, \mu))$ [Hubbard and Masur, 1979] is defined by the real-valued functions v_i on U_i that satisfy a set of conditions.

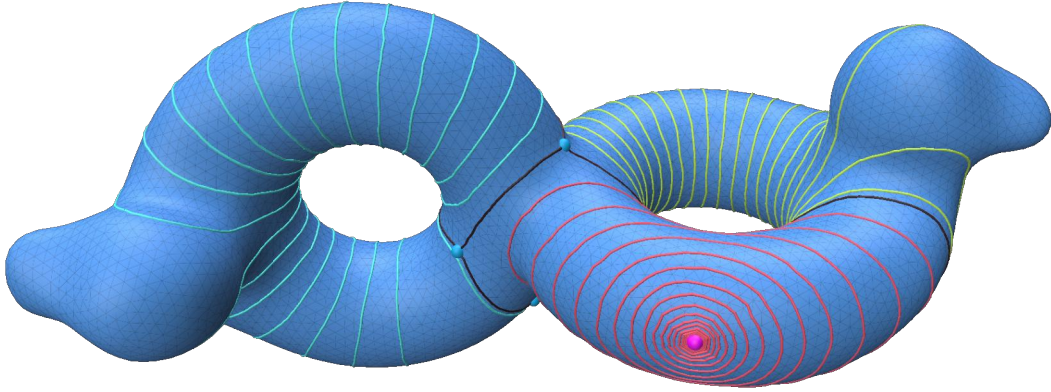


Figure 13: A harmonic foliation with a pole on a two-dimensional discrete mesh, generated by software Geometric.

Definition 3.3 (Whitehead Move [Fathi, 2012]). A Whitehead move is a transformation of one foliation into another, comprising three cases: (1) isotopy; (2) collapsing a finite arc between two zeros; (3) splitting a zero.

Definition 3.4 (Whitehead Class [Fathi, 2012]). Two measured foliations $(\mathcal{F}, \mu)$ and $(\mathcal{G}, \nu)$ are said to be *Whitehead equivalent* if they can be transformed into each other via a sequence of Whitehead moves. All foliations are partitioned into Whitehead classes by this equivalence relation.

Six foliations fall into the same Whitehead class is shown in Fig. 14, they are related to each other by Whitehead move. The non-harmonic foliation in the upper-left picture evolves into the harmonic one in the bottom-right picture.

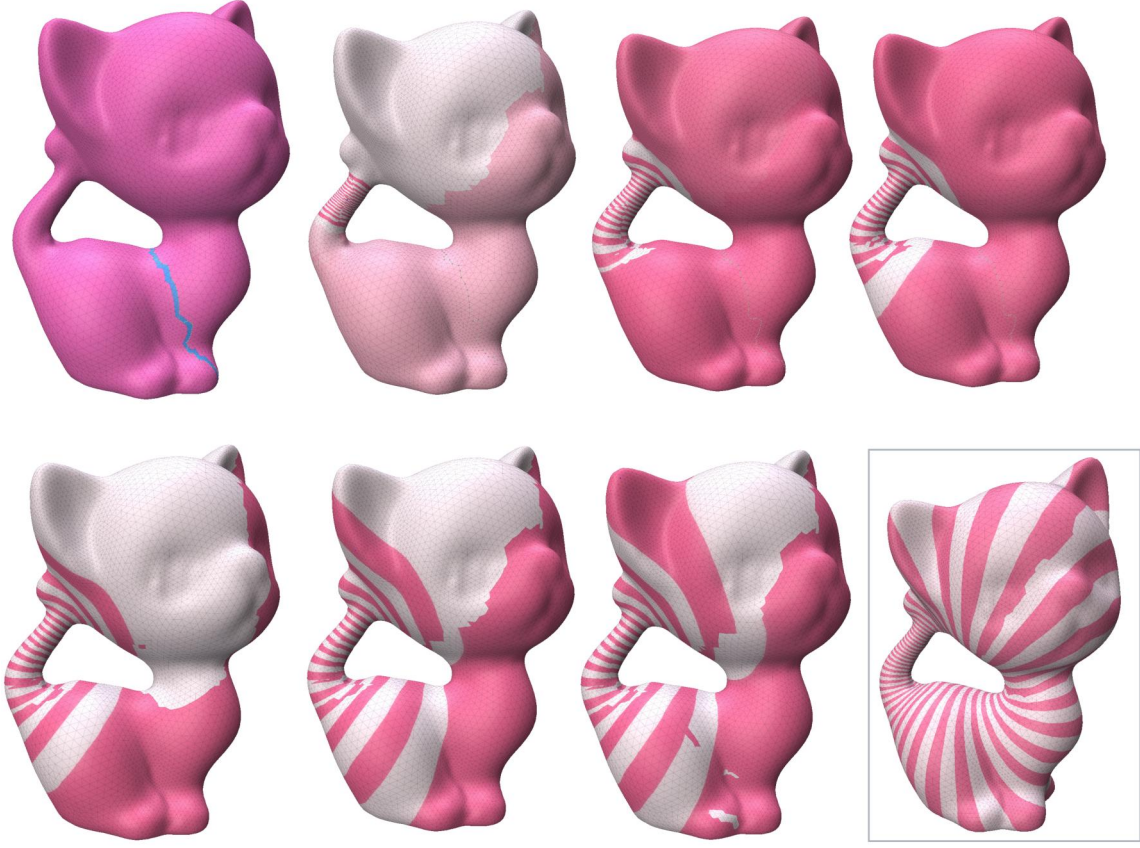


Figure 14: Six foliations belong to the same Whitehead class, generated by software Geometric.

A holomorphic quadratic differential Ω induces a unique harmonic measured foliation within a Whitehead class $\mathcal{F}_\Omega$ (equivalently, it corresponds to a unique *Whitehead class*), and vice versa.

3.5 Dirichlet Energy and Harmonic Measured Foliation

By harmonic analysis [Schoen and Yau, 1997], a harmonic differential 1-form minimizes the Dirichlet energy within its cohomology class. Similarly, as shown in [Gardiner and Lakic, 2010a], there exists a Dirichlet energy D for a measured foliation $|dv|$, and a harmonic foliation uniquely minimizes this Dirichlet energy within its Whitehead class [Gardiner and Lakic, 2010a,b].

There are some facts about measured foliations, holomorphic differential 1-forms and holomorphic quadratic differentials for a closed Riemann surface of genus g .

1. The dimension of the space of all harmonic 1-forms is $2g$.
2. A holomorphic quadratic differential on a genus-zero closed surface must be zero.
3. The complex dimension of the space of all holomorphic quadratic differentials is $3g - 3$, for the genus $g > 1$.
4. A holomorphic quadratic differential has $4g - 4$ *zeros* counting multiplicities.
5. The local structures around a *zero* point of order one are the complex functions: $z \rightarrow z^2$ for holomorphic differential 1-forms and $z \rightarrow z^{\frac{3}{2}}$ for holomorphic quadratic differentials respectively.

4 Discrete Harmonic Foliation

A harmonic foliation on a surface is an abstract mathematical concept representing a global geometric structure. Here, "abstract" means that there are no numerical values directly verifiable on the surface

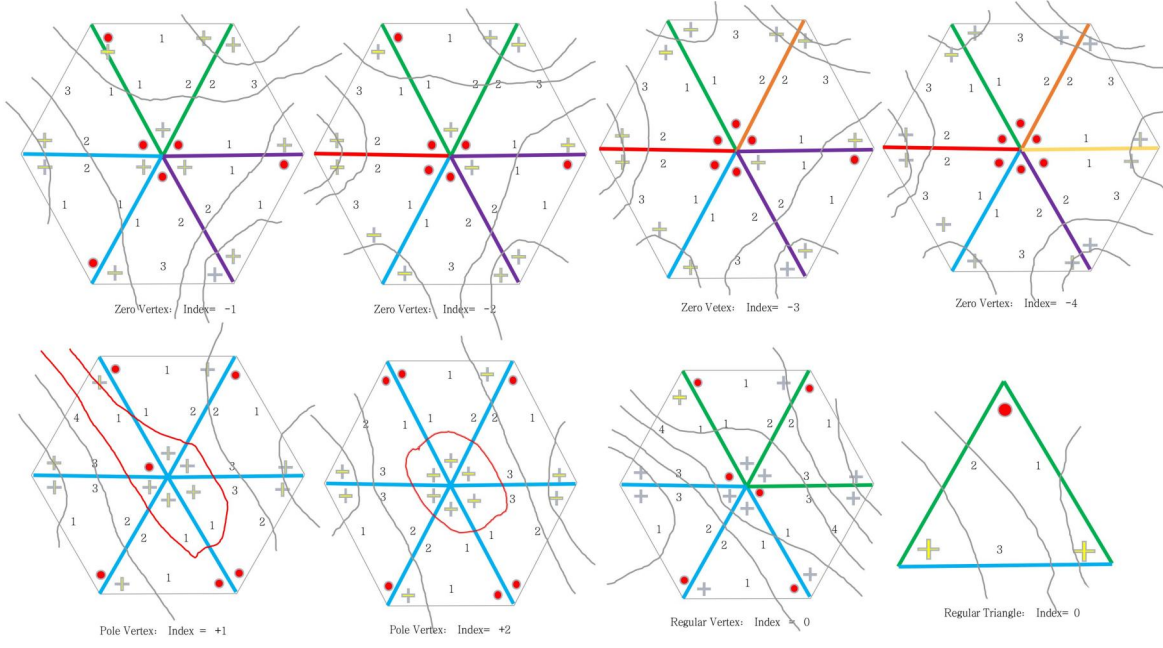


Figure 15: Discrete Whitehead move on a two-dimensional discrete mesh.

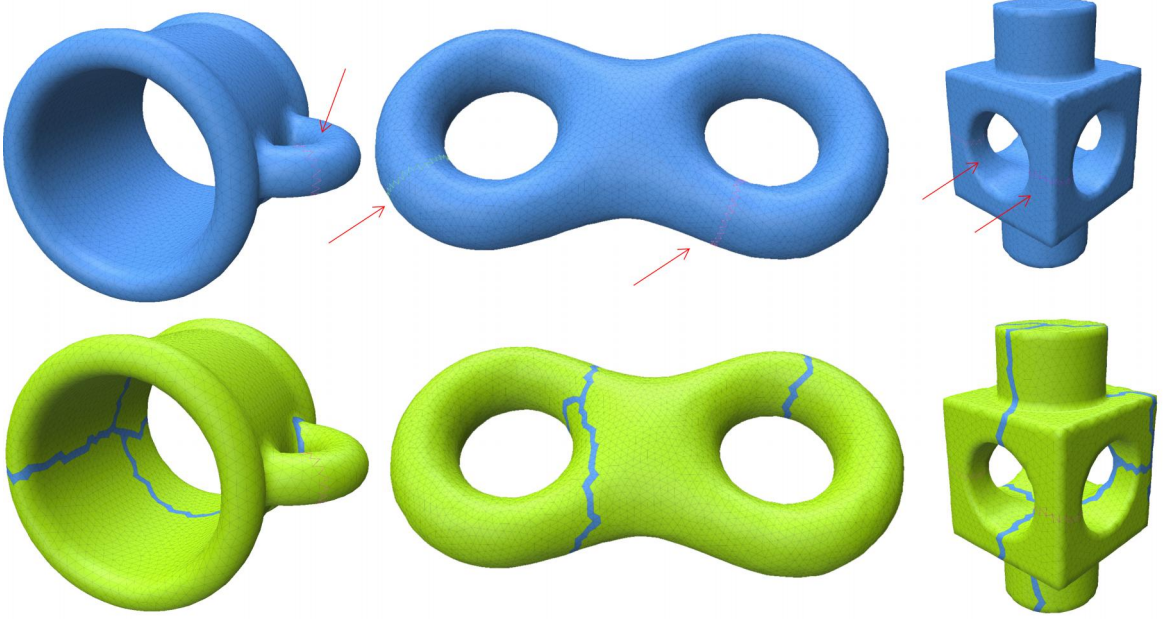


Figure 16: Constrained diffusion of Whitehead move.

itself. Through computer algorithms, an abstract foliation can be converted into a concrete one, allowing numerical values to be assigned to every point on the surface.

A concrete harmonic foliation generated via computer algorithms is analogous to a concrete physical particle, in that both can be examined to obtain specific numerical values through respective methods—algorithmic computation for the former and physical experimentation for the latter.

Drawing on our experience in designing algorithms on meshes [Zhao et al., 2018, 2020b, 2019; Zhao and Gortler, 2016; Zhao et al., 2020a; Zhao, 2016; Zhao et al., 2017], we developed a convergent discrete harmonic foliation algorithm as presented in [Zhao et al., 2022]. This algorithm evolves an input non-harmonic but closed foliation into a harmonic one in the same Whitehead class. There are

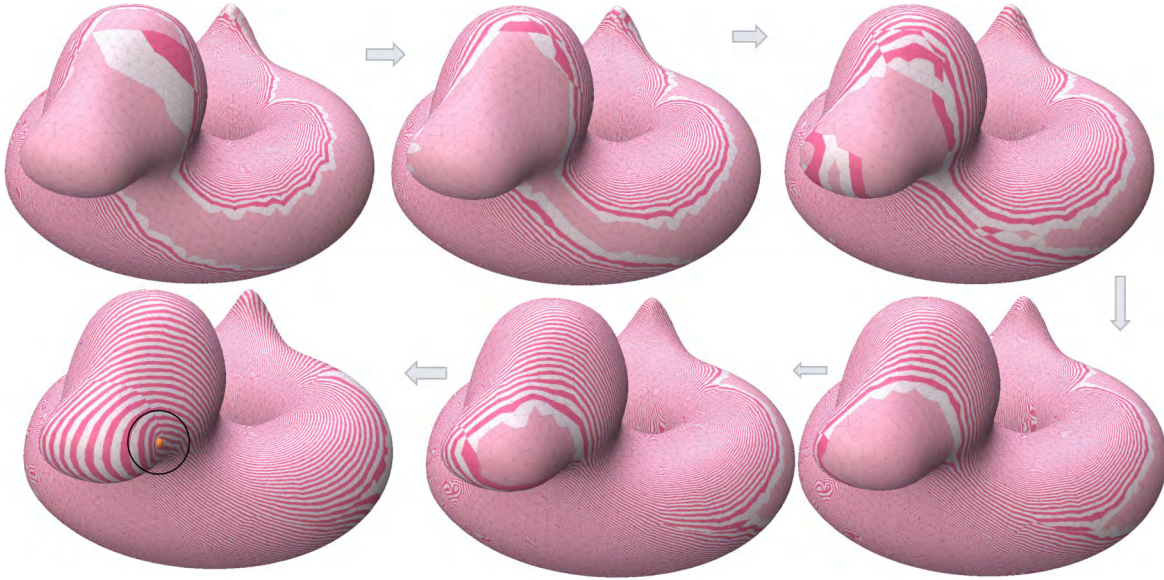


Figure 17: Diffusion of Whitehead move without constraint.

two key steps in evolution: the first is a local discrete Whitehead move applied to each vertex, as illustrated in Fig. 15; the second is the constrained diffusion of Whitehead moves, as shown in Fig. 16. Without constraints, the result will not be a harmonic foliation within the same Whitehead class, as depicted in Fig. 17. Our computational foliations are dynamic, as shown in the [video 3](#), the [video 4](#), the [video 5](#), the [video 6](#).

5 Quantum Mechanics Interpretation of Computational Foliation

Here, we use the term "computational" to distinguish it from an "abstract" foliation: just as nature generates concrete physical particles within its own "natural computer," we construct tangible foliations in a computer via computational algorithms.

The quantum mechanical phenomena exhibited by physical particles arise because these particles are not merely abstract global geometric structures on fiber bundles [Bernstein and Phillips, 1981], they are also concretely constructed by nature itself. These phenomena are well-known and explained in [Sakurai and Napolitano, 2020; Cohen-Tannoudji et al., 2020; Griffiths and Schroeter, 2018; Susskind et al., 2014; Li; Bertlmann and Friis, 2023; Aitchison and Hey, 2021; Dirac, 2001, 2010; Weinberg, 1995; Hall, 2013; Bohm, 1986; Mermin, 2007; Nielsen and Chuang, 2012; Zee, 2023, 2010; Bengtsson and Zyczkowski, 2006]. In this section, we will illustrate certain analogies between quantum mechanical phenomena and the evolution of computational harmonic foliations.

5.1 Wave-particle Duality

Wave-particle duality is a fundamental principle in quantum mechanics that describes the dual nature of all matter and energy: every elementary particle (such as electrons, photons, or even larger entities under certain conditions) exhibits both wave-like and particle-like properties, depending on the experimental setup or context in which it is observed. At the heart of wave-particle duality is the idea that no single classical model—whether purely "wave" or purely "particle"—can fully explain the behavior of quantum-scale entities. Instead, their properties shift based on how they are measured or interacted with.

1. Particle-like behavior: When entities interact with matter (e.g., hitting a detector screen, transferring momentum in collisions), they often behave as discrete, localized "particles." For example:

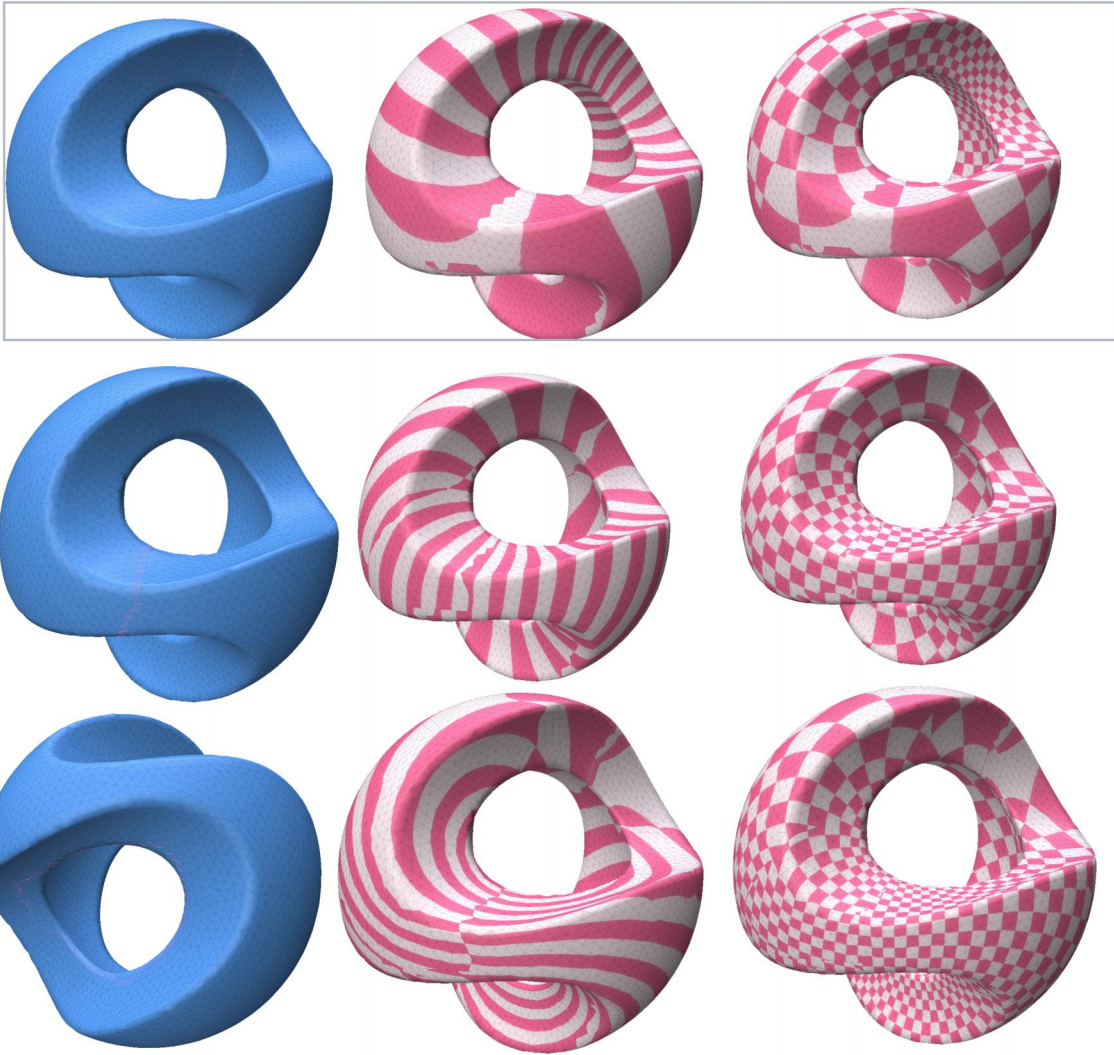


Figure 18: Wave-particle duality of harmonic foliation, generated by software Geometric. The left is three input closed foliation as particles; the middle three are harmonic foliation as waves; the right three are corresponding holomorphic quadratic differentials.

Photons hitting a detector create individual bright spots, like tiny particles. Electrons in a cathode ray tube act as particles, striking pixels to produce light.

2. Wave-like behavior: When entities propagate through space or interact with obstacles (e.g., slits, gratings), they exhibit "wave" properties such as interference and diffraction. For example: In the double-slit experiment, electrons or photons pass through two slits and create an interference pattern on a screen—similar to how water waves overlap to form peaks and troughs. Light diffracts around corners, a classic wave phenomenon.

Wave-particle duality challenges classical intuition but is critical to understanding quantum mechanics. It underlies technologies like electron microscopes using electron waves to image tiny structures and quantum computing which leverages quantum particles' dual properties for information processing. As physicist Richard Feynman noted, the double-slit experiment encapsulates "the only mystery" of quantum mechanics—highlighting how duality defies classical categorization. In short, wave-particle duality is not a contradiction but a reflection of the quantum world's fundamental nature: entities are neither purely waves nor purely particles, but a hybrid that manifests different traits based on observation.

Foliations are classified by their Whitehead classes: all foliations belonging to the same Whitehead

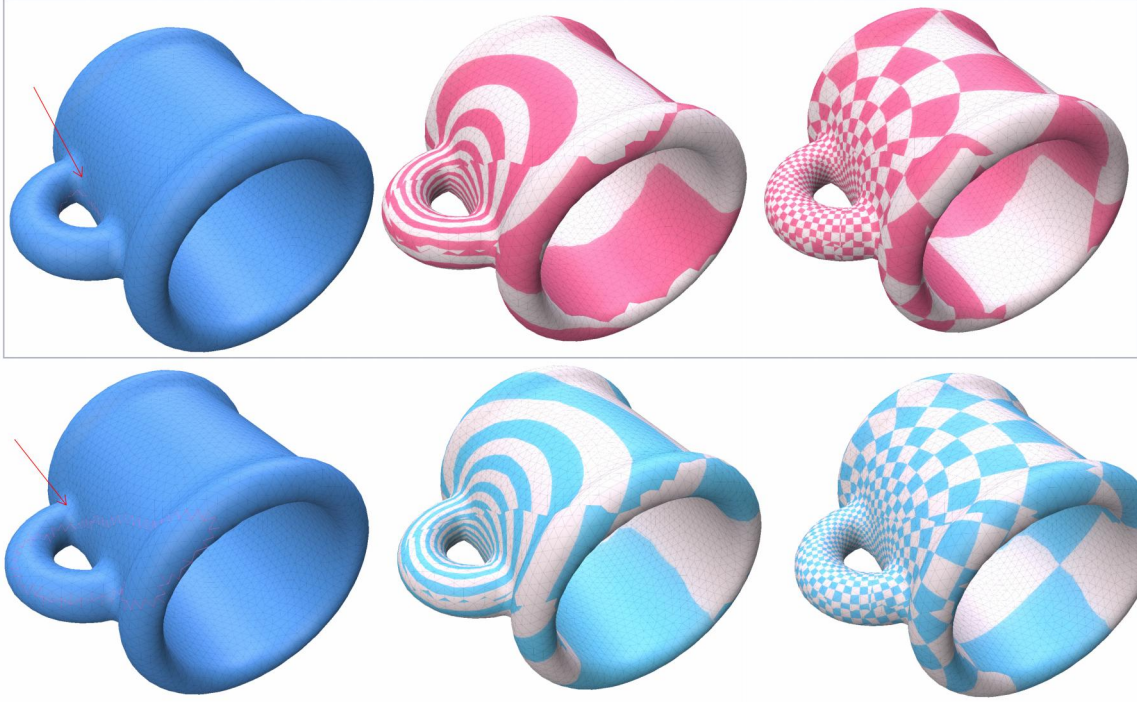


Figure 19: Wave-particle duality of harmonic foliation, generated by software Geometric. The left is two input closed foliation as particles; the middle two are harmonic foliation as waves; the right two are corresponding holomorphic quadratic differentials.

class are regarded as identical objects and can morph into one another. For instance, the input red loop in Fig. 19 represents a foliation, while the harmonic foliation (shown in the middle picture) that diffuses from this input red loop represents another foliation. However, since they belong to the same Whitehead class, the two are considered the same object in the context of Whitehead classes.

From a certain perspective, a computational harmonic foliation also exhibits wave-particle duality:

1. The input foliation, which is closed but not harmonic, represents particles, as shown by the left two pictures in Fig. 19.
2. The foliations arising during the diffusion process of the input foliation denote waves by the middle two pictures in Fig. 19 , video 1 , the video 3 .

The particle aspect of a foliation manifests as a loop—specifically, the red loops in the left-hand images of Fig. 19 on a surface—whereas the particle aspect of a photon is described by a small sphere. These two types of particles exhibit distinct forms because they correspond to different global geometric structures on distinct fiber bundles. However, both types of particles are localized within a specific small region of their respective manifolds.

Actually, a physical photon is not an infinitesimal point; it merely has its maximum value within a small spacetime region. Similarly, our foliation particle has its most value within a small region of a surface. Even though the value of the foliation is zero outside this small region, its global structure is already determined from the start. This structure is preserved by our algorithm; otherwise, the foliation would be destroyed and transformed into another object, as shown in Fig. 17. If its global geometric structure is not preserved, there will be no quantum mechanical analogy.

In string theory, a particle manifests as a string or a Calabi-Yau manifold, thus, the term "particle" does not imply it must take the form of a sphere. Traditionally, wave side and particle side of photon have been regarded as two distinct phenomena. However, through our computational foliations, it can be vividly demonstrated that these two counterparts are, in fact, the same entity, capable of morphing into one another if we look at them from the unified viewpoint of global geometric structures.

The three particle loops in Fig. 18, which are homotopically distinct, represent three distinct loops (or particles) that cannot morph into one another on the surface, as they belong to different Whitehead

classes. In contrast, the two particle loops in Fig. 19 can morph into each other and thus belong to the same particle (i.e., the same foliation Whitehead class).

In the algorithm underlying our computational foliation, the numerical value outside the red loop of a particle foliation on a surface is zero. As a particle foliation diffuses into a wave foliation, the region of the surface with a numerical value of zero enlarges progressively, as shown in Fig. 14 and 18 and the [video 1](#). In this way, our computational foliation undergoes a gradual transformation from its particle aspect to its wave aspect.

The [video 1](#) illustrates the process by which an input particle-loop foliation gradually diffuses into a wave foliation more clearly than the pictures.

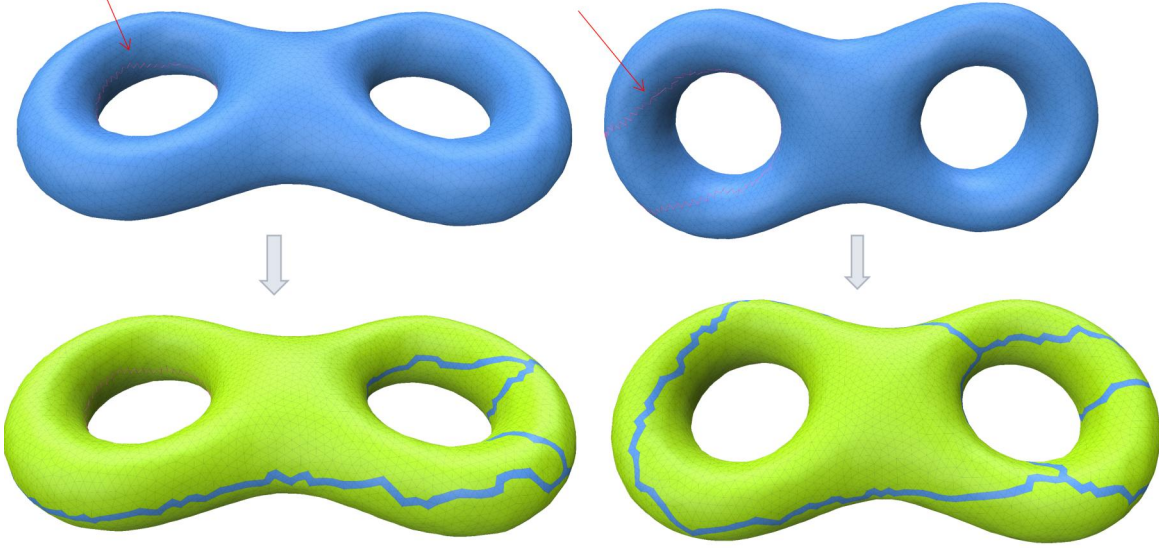


Figure 20: Ribbon graph of a harmonic foliation.

5.2 Quantum Self Interference

Quantum self-interference stands as a remarkable manifestation of wave-particle duality in quantum mechanics, wherein a single quantum particle (such as an electron, photon, or atom) interferes with itself. Unlike classical self-interference, which occurs when multiple waves from the same source overlap, quantum self-interference emerges from a particle's capacity to "exist in multiple states simultaneously" (superposition) and subsequently "interact with itself" when these states recombine. This phenomenon is pivotal to understanding quantum behavior and has been experimentally validated through landmark studies. Superposition refers to the ability of a quantum particle to occupy multiple states (e.g., traverse multiple paths) at the same time. For instance, a photon can "take two paths" through an interferometer concurrently until it is measured. Wavefunction interference implies that a particle's quantum state is described by a wavefunction, which encodes the probabilities of finding the particle in specific states. When the particle's superposed states recombine, their wavefunctions interfere (either constructively or destructively), generating measurable patterns.

Quantum self-interference's experimental examples include the following:

1. **The Double-Slit Experiment** The most renowned demonstration of quantum self-interference: A beam of particles (e.g., electrons or photons) is emitted toward a barrier with two slits. In the absence of a detector at the slits, the particles form an interference pattern on a screen behind the barrier—characterized by bright bands (constructive interference) and dark bands (destructive interference)—as if each particle passed through both slits and interfered with itself. When a detector is added to determine which slit the particle passes through, the interference pattern vanishes, and the particles behave like classical particles (with no self-interference). This indicates that observation collapses the superposition, eliminating the wave-like behavior. As Richard Feynman noted, this experiment is "impossible, absolutely impossible, to explain in any classical way," underscoring its role as a cornerstone of quantum mechanics.

2. **Mach-Zehnder Interferometer (for Photons)** A single photon is split into two paths by a beam splitter. The two paths are then recombined by a second beam splitter. Depending on the lengths of the paths, the photon's wavefunction from each path interferes with itself: Constructive interference: The photon is detected at one output port. Destructive interference: The photon is never detected at the other output port. This demonstrates that even a single photon can undergo self-interference, as its wavefunction "explores" both paths.
3. **Neutron Interferometry** Neutrons, despite their larger mass, also exhibit self-interference: A neutron beam is split into two paths using crystalline structures (which diffract neutrons similarly to how light is diffracted through slits). Recombining these paths results in an interference pattern, confirming that massive particles (not just massless photons) undergo self-interference. Quantum self-interference directly confirms wave-particle duality, a core principle of quantum mechanics that sets it apart from classical physics.

Quantum self-interference enables various technologies, such as:

1. **Quantum computing:** Leveraging superposed states and interference to perform complex calculations.
2. **Precision metrology:** Interferometers used to measure minuscule distances (e.g., gravitational wave detectors like LIGO, which utilize laser self-interference to detect spacetime ripples).
3. **Quantum cryptography:** Secure communication based on the fact that observing a quantum state inevitably disrupts its interference pattern.

It is important to note that quantum self-interference does not involve "two particles interfering"—rather, it involves a single particle interfering with itself due to superposition. Moreover, the interference stems from the particle's intrinsic wave nature, not external interactions.

In summary, quantum self-interference of physical particles is a counterintuitive yet experimentally well-established phenomenon that reveals the wave-like behavior of individual quantum particles. It shapes our understanding of the quantum world and drives innovation in modern technology.

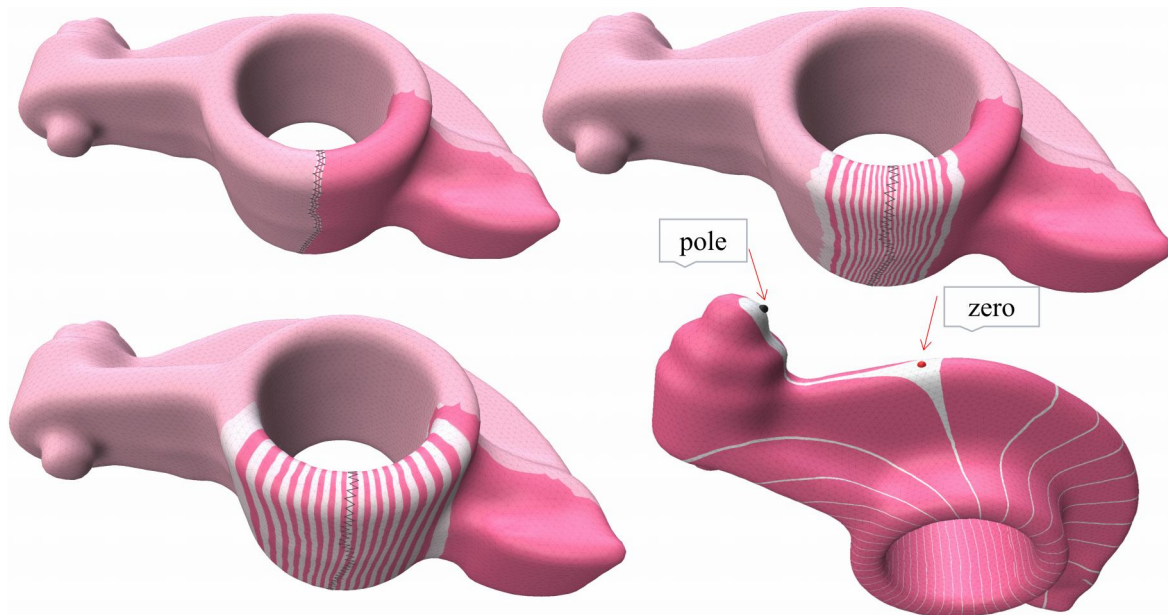


Figure 21: Without self interference of computational foliation, generated by software Geometric.

As a closed foliation evolves across a surface, it will intersect itself during diffusion—giving rise to a phenomenon analogous to the self-interference of physical particles, as illustrated in this [video 1](#) and the Fig. 14. We term this the self-interference of computational foliations. Self-interference in physical particles and computational foliations both are the global property of their geometric structures.

The quantum self-interference of physical particles occurs within linear time and is a construct of nature, which itself could be regarded as a quantum computer. In the case of computational foliations, we simulate the self-interference behavior of foliations in a computer. Notably, this implementation requires a nonlinear algorithm as it is executed on a traditional computer rather than a quantum one. The algorithm for our computational foliation first identifies a structure (as indicated by the blue loops in Fig. 20 and Fig. 16) based on the corresponding input particle foliation (as shown by the red loops in Fig. 20 and Fig. 16) on a surface. A particle foliation ceases diffusion once its boundary forms a ribbon graph. Diffusion then occurs within this ribbon graph, and upon completion, resumes across the entire surface.

Self-interference is crucial to computational foliations; without it, the evolution of the foliation would fail. For instance, as illustrated in Fig. 21, an input particle foliation without pole would transform into another foliation with a pole. These two foliations belong to different Whitehead classes and are not the same geometric object.

If a concrete foliation exists in nature, its self-interference would occur spontaneously without the aid of a ribbon graph. In our case, since we are simulating such self-interference without relying on a quantum computer, we require certain algorithms running in a traditional computer to assist in preserving its global structures.

5.3 The Schrödinger Equation

The Schrödinger equation is a fundamental equation in quantum mechanics that describes how the quantum state of a physical system evolves over time. It was formulated by the Austrian physicist Erwin Schrödinger in 1925, laying the mathematical foundation for quantum theory alongside other key developments like matrix mechanics. In classical mechanics, the motion of objects is governed by Newton's laws, which predict precise positions and velocities. In quantum mechanics, however, particles and systems exhibit wave-particle duality, and their behavior is probabilistic. The Schrödinger equation captures this wave-like nature, enabling predictions about the likelihood of finding a particle in a specific state or location at a given time.

The Schrödinger equation describes how the wave function $\psi(\mathbf{r}, t)$ evolves with time. Its general form is:

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \hat{H} \psi(\mathbf{r}, t)$$

where:

1. - $i = \sqrt{-1}$ (imaginary unit),
2. - $\hbar$ (reduced Planck constant) $\approx 1.054 \times 10^{-34}$ J·s,
3. - $\frac{\partial \psi}{\partial t}$ is the partial derivative of ψ with respect to time t ,
4. - $\hat{H}$ is the **HAMILTONIAN OPERATOR** (total energy of the system: kinetic + potential),
5. - $\psi(\mathbf{r}, t)$ is the wave function (depends on position $\mathbf{r}$ and time t).

The wave function ψ itself has no direct physical meaning, but the square of its magnitude, $|\psi|^2$, represents the probability density of finding the particle at a specific position $\mathbf{r}$ and time t . This probabilistic interpretation, developed by Max Born, is a cornerstone of quantum mechanics.

In our computational foliation, the numerical values of every point on the whole surface also evolve from a particle foliation to a wave foliation, following a rule that satisfies a harmonic function constraint [Zhao et al., 2022], as shown in Fig. 15 and the [video 1](#), the [video 2](#). The value at each point gradually converges to a harmonic function over the course of evolution by a sequence of discrete Whitehead moves:

$$\Delta u = 0 \tag{1}$$

The role of diffusion in computational foliations is analogous to that of the Schrödinger equation in physical particles. As they are two different global geometric structures, thus computation foliation is not expected to have the exact same evolution formula as the Schrödinger equation. In foliations, the numerical values are real; however, the corresponding holomorphic quadratic differentials of the foliations take complex values. The square of the magnitude of these differentials can be interpreted as their probability density, analogous to the probability density in quantum particles.

It is worthwhile to explore how to unify the Schrödinger equation and the evolution of computational foliation within a single framework.

5.4 Entanglement of Two Foliations

Quantum entanglement is a fundamental phenomenon in quantum mechanics wherein two or more particles become correlated such that the quantum state of each particle cannot be described independently of the others—even when the particles are separated by vast distances. This correlation persists irrespective of the spatial gap between them, defying classical intuition.

Key characteristics of quantum entanglement:

1. **Non-locality:** For entangled particles, measuring a property (e.g., spin, polarization) of one instantly determines the corresponding property of the other, regardless of how far apart they are. This phenomenon, famously criticized by Einstein as "spooky action at a distance," has been experimentally confirmed through tests of Bell's inequalities.
2. **Shared Quantum State:** Entangled particles are characterized by a single joint quantum state rather than individual states. For instance, two entangled electrons might exist in a state where "if one has spin-up, the other must have spin-down"—yet neither possesses a definite spin until measured.
3. **Irreducibility:** Entanglement cannot be reduced to classical correlations. Unlike two pre-set coins (one heads, one tails), entangled particles have no predetermined properties; their states are defined solely through their mutual connection.

A well-known illustration is the Bell state involving two photons with entangled polarizations: If photon A is measured to have vertical polarization, photon B will instantly be found to have horizontal polarization, and vice versa. This relationship holds even if the photons are light-years apart at the time of measurement.

Quantum entanglement forms the foundation of emerging technologies such as quantum computing (enabling parallel processing via entangled qubits), quantum cryptography (facilitating unhackable communication through entangled particles), and quantum teleportation (transferring quantum states using entanglement). It also challenges our understanding of reality, underscoring the profound distinctions between quantum and classical physics. As physicist John Bell noted, entanglement demonstrates that quantum mechanics is either non-local or incomplete—and decades of experiments strongly support the former.

Quantum entanglement emerged from debates in the 1930s regarding the "completeness" of quantum mechanics:

1. **The Einstein-Podolsky-Rosen (EPR) Paradox (1935):** Einstein and his colleagues argued that quantum mechanics must be incomplete. They proposed a thought experiment: two particles interact and then separate. If entangled, measuring a property (e.g., position) of one instantly reveals the property of the other—even across great distances. Einstein termed this "spooky action at a distance," suggesting that "local realism" (the idea that particles have pre-existing properties and no signal travels faster than light) must explain the correlation.
2. **John Bell's Theorem (1964):** Bell derived a mathematical inequality showing that no local realist theory could replicate all predictions of quantum mechanics. Experimental violations of Bell's inequality would confirm the existence of entanglement and quantum non-locality.
3. **Experimental Confirmation:** By the 1980s (in experiments by Alain Aspect) and further rigorous tests in the 2020s, Bell's inequality was repeatedly violated, confirming that entanglement is real and non-local.

Contrast with classical systems:

1. **Classical systems** (e.g., two coins, two billiard balls) adhere to local realism: Objects possess definite properties before measurement (e.g., a coin is "heads" even if unobserved). Interactions rely on signals traveling no faster than light.

2. Entanglement violates both principles: No Pre-Existing Properties: Entangled particles have no fixed properties until measured. The singlet state (e.g., for electrons) does not "hide" pre-determined spin values; it merely defines a correlation that emerges upon measurement.
3. Non-locality: Measuring one particle instantly influences the other, regardless of distance. Importantly, this is not "communication" (no usable information is transmitted faster than light) but a fundamental connection encoded in their shared quantum state.

Entanglement arises when particles interact and share a quantum state. Common mechanisms include:

1. Particle Decay: A subatomic particle (e.g., a pion) decays into two photons, which become entangled in polarization.
2. Photon Interactions: Lasers passing through certain crystals (e.g., beta-barium borate) can split photons into entangled pairs.
3. Quantum Gates: In quantum computers, operations like the CNOT gate entangle qubits (quantum bits).



Figure 22: Quantum entanglement imitation of two computational foliations. The up right picture is two input particle foliations; the bottom left picture is their entanglement, which is a new foliation.

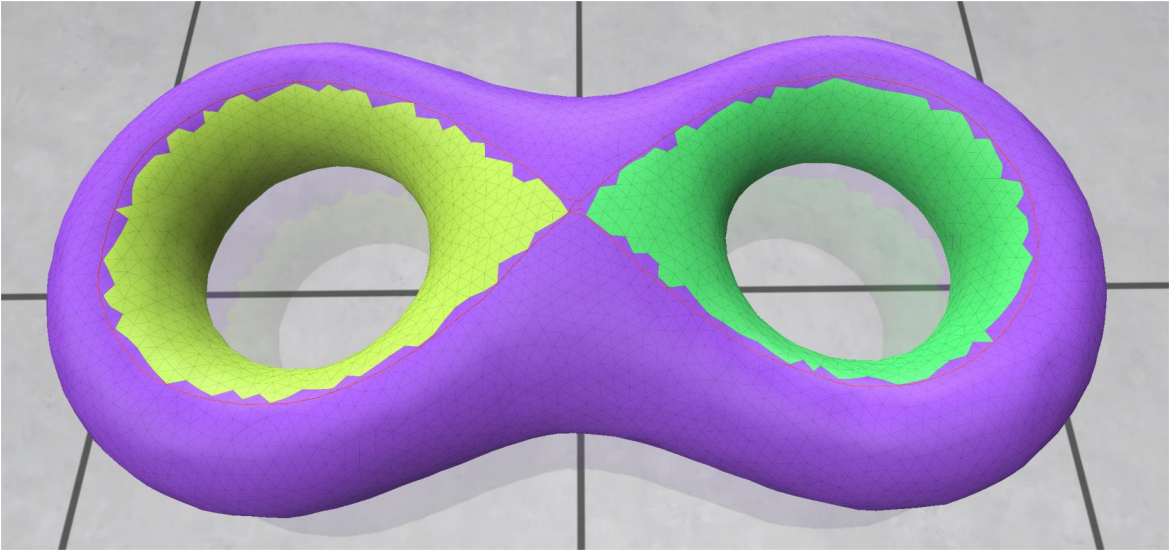


Figure 23: The diffusion of two computational foliations on the same surface.

If two computational foliations evolve independently, as shown in Fig. 18, they will not become entangled. However when two computational foliations evolve on the same surface, as illustrated in

Fig. 22, Fig. 23, the [video 3](#), the [video 4](#), the [video 5](#), they can exhibit phenomena analogous to quantum entanglement. The details depicted in Fig. 22 are as follows:

1. The red loops in the upper-right panel represent the two input particle foliations.
2. These two foliations evolve on the same surface in accordance with the rule proposed in [Zhao et al., 2022], as shown in the upper-middle, upper-right, and lower-right panels of Fig. 22.
3. In the lower-middle panel, the two foliations nearly intersect and form a ribbon graph; however, they are not yet entangled at this stage.
4. Finally, in the lower-left panel, the two foliations merge into a third foliation, giving rise to an entanglement. This behavior is analogous to quantum entanglement.

Once entangled, the two foliations become inseparable from one another. If we were two-dimensional beings living on a surface, we would only be able to observe the two foliations independently from our local perspective. Fortunately, as we inhabit a four-dimensional spacetime, we can adopt a global perspective of the computational foliations beyond their two-dimensional surface, thereby making the entanglement of the two foliations manifest and be apparent.

Quantum entanglement is constructed by nature itself which has a mechanics of in-built global geometric structure with line time. In contrast, our computational foliations achieve entanglement from its global geometric structure through algorithms executed on traditional computers, which function in non-linear time. As a result, our approach incurs a cost in terms of runtime.

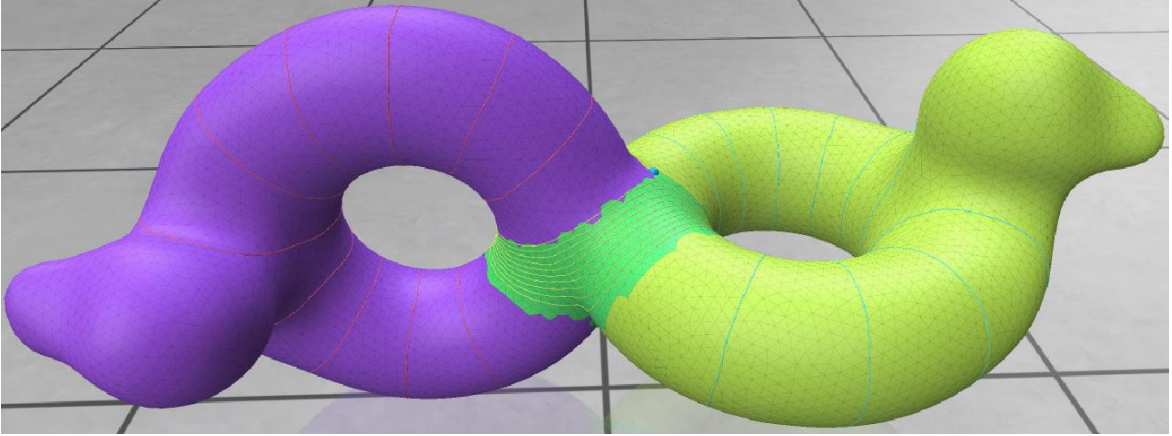


Figure 24: The 90 percent value of a foliation is on blue region in the middle of picture.

If we adjust the diffusion process of computational foliations such that 90 percent of the numerical value remains in the input loop foliation, with only the remaining 10 percent spreading to other regions of the surface (as shown in Fig. 24, the result will more closely resemble the entanglement of two particles. No matter how less value are diffused, once they are merged, they become a new foliation.

From this, we can discern the fundamental reason behind the quantum entanglement of two physical particles: all counterintuitive phenomena arise because we, as three-dimensional beings, observe these three-dimensional physical particles from a local perspective.

It will also be interesting to explore how to design an algorithm for our computational foliations to simulate results analogous to quantum entanglement decoherence in the future, which separates the new third foliation into the original two.

5.5 Quantum Tunneling of Computational Foliations

Quantum tunneling is a fundamental phenomenon in quantum mechanics wherein a particle can traverse a potential energy barrier even when its total energy is less than the barrier's height—an event that would be impossible under the framework of classical physics. In classical mechanics, a particle with insufficient energy to overcome a barrier would simply rebound. Quantum mechanics, however,

introduces the concept of wave-particle duality: particles can be described by wavefunctions, which encode the probability of detecting the particle in a specific state or position. When a quantum particle encounters a potential barrier, its wavefunction does not abruptly vanish at the barrier's boundary. Instead, it decays exponentially as it penetrates the barrier, meaning there exists a non-zero probability that the particle will "tunnel" to the other side—even without sufficient energy to surmount the barrier in a classical sense.

Key Features of quantum tunneling:

1. Probabilistic Nature: Tunneling is inherently probabilistic. The likelihood of a particle tunneling depends on factors such as the barrier's height, width, and the particle's energy. Lower, narrower barriers increase the tunneling probability.
2. Microscopic Scale: It is most prominent for subatomic particles (e.g., electrons, protons) due to their small mass, which amplifies the significance of their quantum wave-like properties.
3. Practical Applications: Quantum tunneling enables technologies such as scanning tunneling microscopes, which map atomic surfaces by detecting electrons tunneling between a probe and the surface. It also plays a critical role in nuclear fusion (e.g., within stars), semiconductor devices, and quantum computing.

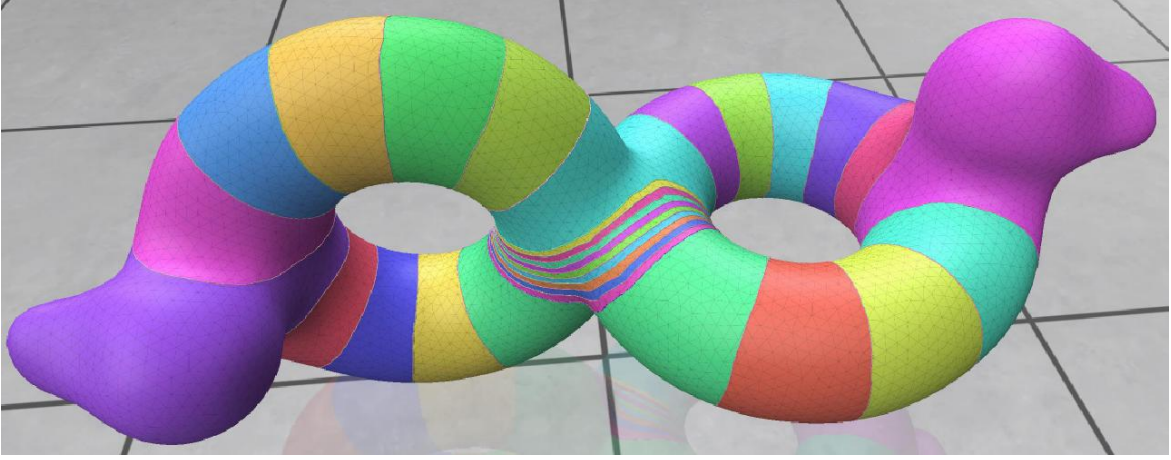


Figure 25: The region in the middle contains most of value of a foliation.

This phenomenon challenges intuitive classical notions of motion and underscores the unique, counterintuitive behavior of matter at the quantum level. At the quantum scale, particles (such as electrons or photons) exhibit not only discrete "particle-like" behavior but also wave-like properties—a concept known as wave-particle duality. This wave nature is described by a mathematical function called the wavefunction, which encodes the probability of finding the particle at a specific position or in a specific state. When a quantum particle encounters a potential energy barrier, its wavefunction does not abruptly vanish at the barrier's edge. Instead: A portion of the wave reflects off the barrier. A portion of the wave penetrates the barrier, decaying exponentially as it propagates through. If the barrier is sufficiently thin, this decaying wavefunction can still reach the opposite side—indicating a non-zero probability that the particle has "tunneled" through, even when its energy is less than the barrier's height.

In our computational foliation model, we can designate a specific region on the surface to contain 99 percent of the foliation's value, with only 1 percent remaining outside this region, as shown in Fig. 25. Subsequently, the evolution of the foliation can be constrained within this region. Within the region, the foliation behaves like a particle. Nevertheless, the foliation can still manifest as the same particle outside this region. This is similar in spirit with Quantum tunneling of physical particle.

Quantum tunneling is a manifestation of the global geometric structure within four-dimensional spacetime. Psychologically capturing this phenomenon is challenging, as we observers exist within spacetime itself. However, through computational foliations on a two-dimensional surface, the underlying mechanism of quantum tunneling becomes far more evident.

5.6 Computational foliations on a Surface with Black Hole

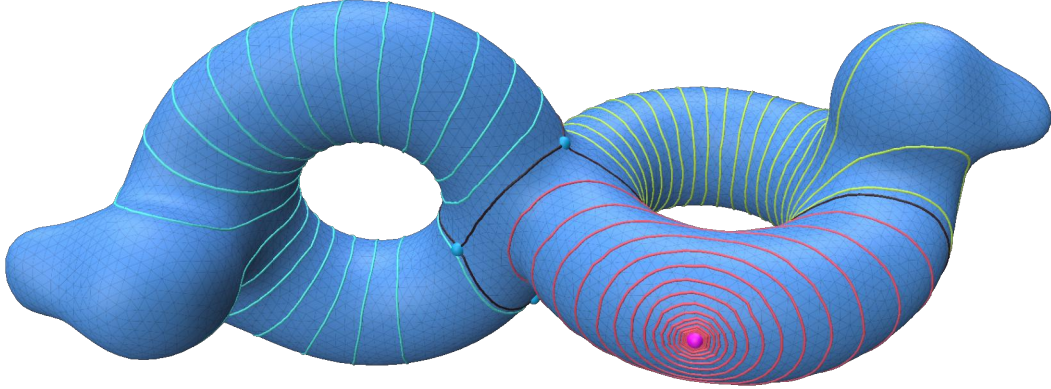


Figure 26: A computational foliation on a surface with a "black hole".

A global geometric structure, such as a physical particle or a computational foliation, resides on an entire manifold, even though it only takes non-zero values within a local region. They can sense the entire universe or the entire surface, respectively, from a local region. After evolution, a photon or a foliation will come to cover the entire manifold.

If there is a black hole in spacetime, a photon, as a global geometric object, will adjust to it automatically. A computational foliation will behave likewise.

A "black hole" in a surface of genus g is a puncture. A foliation has a "pole" with degree 2 on the puncture. The number of "zeros" and "poles" satisfies the rule:

$$\text{zeros} \times \text{degree} - \text{poles} \times \text{degree} = 4g - 4 \quad (2)$$

A computational foliation on it is shown in Fig. 26. The pink sphere represents a "black hole" or a "pole", the blue spheres denotes "zeros". There are one "pole" of degree 2 and six "zeros" of degree 1.

A question arises here: Given that geometric structures depend on the topology of the manifold or fiber bundle, what happens to a geometric structure if the manifold undergoes a topological change after the structure has formed? It would transform into another global geometric structure. But it is still a global object.

6 Conclusion and Future Work

In this paper, we suggest an explanation of quantum mechanics using the global geometric structures living on fiber bundles. The quantum mechanics of physical particles is formulated within a fiber bundle over the four-dimensional spacetime manifold.

All counterintuitive phenomena arise as manifestations of its global structure. However, it is impossible to observe four-dimensional spacetime in its entirety, only local observations are feasible.

We then introduce another global geometric structure: a computational foliation residing on a two-dimensional surface, which can be examined both locally and globally. By drawing analogies between these structures, we can thus gain a deeper and more comprehensive understanding of the quantum mechanics of physical particles through the visual and algorithmic demonstrations of our computational harmonic foliation. In other words, we simulate quantum mechanical phenomena using another global geometric structure on a traditional computer, but this comes at the cost of employing nonlinear time algorithms—unlike the linear time required in a quantum computer.

The power of quantum computing and quantum computers [Preskill, 2021] stems from the properties of the global geometric structures of physical particles. Consequently, quantum computers need not be dependent on specific physical particles such as photons or electrons. It is possible to develop

quantum computers using other global geometric structures, provided that these structures can be physically constructed. If such new physical global geometric structures reside not in the full four-dimensional spacetime but in two-dimensional or three-dimensional materials—for instance, Majorana modes in iron-based superconductors [Zhu et al., 2020; Lv et al., 2021; Ma et al., 2018; Ding et al., 2008], then quantum computers built from them should be easier to construct, and these structures would exhibit far greater stability in maintaining entanglement to avoid complicated quantum error correction.

The advantage of quantum computers over traditional ones lies in the fact that certain nonlinear operators in traditional computers become linear in quantum computers. The evolution of our computational harmonic foliation is nonlinear when implemented on a traditional computer. If a quantum computer were available, it would potentially enable a linear computational harmonic foliation.

Quantum computers based on other potential global geometric structures should go beyond bits in “qubits”; they could even use “quantum-five” or others, up to specific configurations determined by the global geometric structures, provided that we can physically manipulate these materials. If several global geometric structures are physically realizable in materials, in future, quantum computers may well have more than one kind of architecture.

Our computational algorithm in a sense can be regarded as a “black box” that simulates a quantum computer at the cost of non-linear time to preserve the global structure. A quantum computer is essentially a machine capable of manipulating a type of geometric structure, which can be realized by nature itself to preserve the global structure.

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